HACK4HUMANITY 2026 · ROUND 1 SOLUTION DOCUMENT

Track C — AI for Societal Good | IEEE JCTS · IEEE SIGHT · BRAIN Foundation, Pune

KisaanRaksha

किसानरक्षा

Farmer Financial Distress Early Warning System — Maharashtra

An AI agent that predicts farmer financial crisis before it happens, alerts officers proactively, and helps farmers file insurance claims in Marathi over WhatsApp voice.

SDG 2 · Zero Hunger **SDG 3 · Good Health & Well-Being** **SDG 10 · Reduced Inequalities**

Team: Sahil Karande · Pranav Shripannavar

GitHub: github.com/sahilkarande0918-cmd/Kisan-Raksha-project

July 2026

Contents

1. The Problem — A Farmer Dies Every 3 Hours

Maharashtra loses a farmer to suicide every 3 hours (2025 state data). In 2024, **2,706 farmers died in Vidarbha and Marathwada alone** — and of those cases, only 1,563 were even deemed eligible for government aid.

The system's failure is structural: every rupee of state response is reactive — ex-gratia compensation paid to a family after a death. There is zero predictive infrastructure. Yet the crisis that ends in suicide follows a well-documented, measurable pattern that builds over months:

- A failed monsoon (rainfall deficit) destroys the kharif crop,
- mandi prices fall below MSP just when the farmer must sell,
- and the crop-loan repayment deadline arrives with nothing to pay it.

Every one of those signals is publicly observable in real time — rainfall from weather archives, prices from Agmarknet, crop health from satellites, repayment windows from the crop calendar. **The data to see the crisis coming already exists. Nobody is looking.**

1.1 Our north star beneficiary

A cotton farmer in Amravati taluka with 2 acres, a ₹1.2 lakh crop loan, a feature phone running WhatsApp, and Marathi as his only fluent language. Every design decision in KisaanRaksha traces back to whether it works for him: voice-first (literacy-independent), Marathi-first, WhatsApp (no new app), and offline-tolerant infrastructure on the government side.

2. The Solution — Predict, Alert, Assist

KisaanRaksha is an agentic AI early-warning system with three jobs:

- **PREDICT —** a LightGBM model fuses four live signals into a Financial Stress Index (FSI, 0–100) per district. FSI > 75 = critical.
- **ALERT —** when FSI crosses critical, the duty agriculture officer receives a WhatsApp alert with the evidence: deficit %, price gap, satellite NDVI, farmers in repayment window — before a crisis, not after a death.
- **ASSIST —** the farmer talks to the system in Marathi voice notes on WhatsApp; it answers in Marathi with grounded numbers and auto-drafts his PMFBY crop-insurance claim letter, evidence attached.

16 districts covered (Vidarbha + Marathwada)	**4** grounded live signals fused per district	**87.9** FSI in labeled crisis simulation → alert + claim fired on a real phone

3. Architecture — Custom MCP Server, No Bare LLM Calls

The system implements the track's reference pattern (onboarding slide 7) end-to-end: every fact the agent states comes from a role-scoped MCP tool grounded in a real dataset — the LLM never answers from its training data.

End User Farmer — WhatsApp Marathi voice	→	**Agent / LLM** tool-calling loop (Groq Llama-3.3 / Gemini)	→	**MCP Server** FastMCP — 7 role-scoped tools	→	**Grounded Data** Open-Meteo · Agmarknet · NASA MODIS	

Inbound: Twilio WhatsApp Sandbox → FastAPI webhook → Whisper large-v3 ASR (Marathi/Hindi auto-detect) → agent. Outbound: Twilio sender for farmer replies, officer alerts, and claim letters. A Streamlit choropleth dashboard gives district officers the live FSI heatmap.

3.1 The seven MCP tools

Tool	One clear job	Grounding
query_weather_signal	30-day rainfall deficit vs 5-yr baseline → drought signal	Open-Meteo ERA5 archive
query_mandi_prices	median mandi price vs MSP → price-stress signal	Agmarknet via data.gov.in
query_ndvi_signal	satellite crop-health → vegetation-stress signal	NASA MODIS MOD13Q1 (ORNL)
compute_fsi	fuse 4 signals through LightGBM → FSI 0–100	trained model, held-out validation
get_farmer_history	structured memory across sessions (hashed phone key)	SQLite — no raw PII
send_officer_alert	WhatsApp alert with evidence when FSI critical	Twilio
draft_pmfby_claim	Marathi PMFBY letter filled with the actual evidence numbers	LLM + FSI evidence

Each tool has one job (track slide 9: role-scoped agents); the agent's decision chain is fully traceable through logged tool calls; tools are auditable and swappable — another team could reuse this MCP server as-is.

4. Technical Methodology

4.1 Four signals → Financial Stress Index

Signal	Source (live API)	What it measures
Drought probability	Open-Meteo ERA5 (keyless)	30-day rainfall vs same window averaged over previous 5 years
Mandi price deviation	Agmarknet, data.gov.in	% gap of median modal price below MSP (2025-26 CACP)
NDVI satellite index	NASA MODIS MOD13Q1 via ORNL	actual crop canopy health from space, 16-day composites

Signal	Source (live API)	What it measures
Repayment proximity	district crop calendar	nearness to the crop-loan due window — the coincidence amplifier

4.2 The FSI model — and what its metrics do and do not mean

A LightGBM regressor maps the four signals plus month/crop/region context to FSI 0–100. It is trained on our custom 13,644-row synthetic-but-grounded dataset (Section 6) and evaluated on **held-out talukas the model has never seen**.

Consistency check on synthetic data (held-out talukas)	Value
Mean absolute error (0–100 scale)	3.75
R^2	0.942
Critical-alert recall (FSI > 75)	0.78
Critical-alert precision	0.83

What these numbers mean — stated plainly: the synthetic labels are generated by our own domain-logic formula (weighted signals + coincidence interactions + noise), so on this data the model is recovering a function we designed. These metrics therefore measure **internal consistency of the pipeline — that the model faithfully encodes the domain logic and generalizes it across unseen talukas — not real-world predictive power.** Real validation is exactly what the pilot in Section 8 is designed to produce: alert precision measured against officer-verified ground truth over one monsoon season. We consider it a feature of this submission that the ML claim is scoped honestly.

The label formula itself encodes the core domain insight: distress spikes when crop failure coincides with the loan-repayment deadline — interaction terms (drought × repayment, price × repayment) amplify the base signal, mirroring the documented Oct–Jan clustering of farmer suicides.

Why FSI > 75 as the critical threshold? It is a tunable operating point, not a discovered truth: 75 balances alert recall (0.78) against precision (0.83) on the synthetic distribution, i.e. roughly one false alert per four raised. In deployment the threshold is calibrated per district against officer response capacity during the pilot.

4.3 Context engineering (track slide 8, all six elements)

Track requirement	KisaanRaksha implementation
Tool-grounded reasoning	agent must call MCP tools for every fact; zero bare LLM answers
Structured memory	SQLite farmer history: district, crop, past FSI, claims — recalled every turn
Localization-aware context	Marathi-first replies; Hindi/English auto-detected and matched; MSP in ₹/quintal; correct Devanagari district spellings pinned
Retrieval-augmented grounding	crop calendar, MSP tables, district registry injected as curated context

Track requirement	KisaanRaksha implementation
Guardrails & validation	FSI thresholds decide alerts (not the LLM); simulation mode always labeled; helpline number in critical replies; no promises of money
Multi-step planning	plan → signals → FSI → alert decision → claim draft, state carried across steps

5. Responsible AI — §8.3 Compliance & Bias Audit

5.1 Fairlearn bias audit (mandatory for vulnerable-population models)

The FSI drives officer attention toward or away from farmers, so we audit the critical-alert decision (FSI > 75) on held-out talukas. The harm metric — and the one we gate on — is the false-negative rate: a missed alert is a missed farmer.

Group	Alert rate	False-negative rate	MAE
Vidarbha	0.064	0.205	3.74
Marathwada	0.077	0.229	3.75
Gap across regions	0.013	0.024	0.015
Gap across crops (cotton/soy/tur)	0.045	0.074	0.23

Fairness gate	Threshold	Observed	Result
FNR gap — region	< 0.10	0.024	PASS
FNR gap — crop	< 0.10	0.074	PASS
MAE gap — region	< 2.0 points	0.015	PASS
MAE gap — crop	< 2.0 points	0.23	PASS
Alert-rate gap	monitored, not gated	0.013 / 0.045	—

Gates apply per metric: false-negative-rate gaps are gated at 0.10; MAE gaps (0–100 scale) at 2.0 points. The alert-rate difference is monitored but deliberately not gated — Marathwada's higher alert rate reflects its genuinely higher drought propensity, and equalizing it would suppress true alerts there. Full report: model/fairlearn_report.md, regenerated by model/fairlearn_report.py on every retrain.

5.2 Data ethics

- No real farmer PII anywhere: training data is synthetic; live farmers are keyed by hashed WhatsApp numbers.
- Every data source cited in README: Open-Meteo (CC-BY 4.0), Agmarknet/data.gov.in, NASA MODIS, CACP MSP, public GeoJSON boundaries.

- Demo crisis scenario is always labeled 'SIMULATION' in every message and dashboard view — no fabricated live claims.

5.3 Honest limitations

- As stated in §4.2: the model's metrics measure internal consistency on synthetic data whose labels we generated — they are not evidence of real-world predictive power. The pilot (§8) exists precisely to produce that evidence before any resource-allocation use.
- District-level signals mask farm-level variance; NDVI at district HQ is a proxy, not a per-field measurement.
- The model flags financial stress patterns — it is an officer-attention tool, never an automated eligibility or denial decision.
- Mandi price coverage is thin off-season; the pipeline discloses its fallback scope (district → state → national → seasonal baseline) in every response.

6. Custom Dataset — Flagged for IEEE Dataport

maharashtra_ag_stress_dataset.csv — 13,644 rows of (district × taluka × crop × year × month) panels for 16 Vidarbha/Marathwada districts, with the four stress signals and a documented financial-stress label. Fully synthetic (zero PII), fully reproducible (seeded generator script), and grounded in real distributions: IMD regional drought propensity, observed 2024–26 Agmarknet-vs-MSP gaps, MODIS kharif NDVI seasonality, and real CACP MSP values.

Per the bonus pathway (onboarding slide 11), we flag this dataset for organizer-supported publication on IEEE Dataport. Generation methodology: dataset/README.md.

7. Impact — Accessibility & Scalability

7.1 Accessibility (works within real Indian constraints)

- Zero new app, zero literacy requirement: WhatsApp voice in the farmer's own language (Marathi-first; Hindi/English auto-matched).
- Works on any phone that runs WhatsApp; the farmer-side cost is one voice note.
- Offline-first server design: every signal falls back to last-known cached values, so the dashboard and FSI survive connectivity loss (track slide 13).
- Evidence-based problem: 2,706 deaths (2024, Vidarbha+Marathwada), 42% deemed ineligible for aid — cited state data, not a hypothetical persona.

7.2 Scalability (1 taluka → state footprint)

- All data sources are pan-India government/public APIs — adding a district is one JSON entry, no new data engineering.
- Cost-aware: free-tier LLM inference, keyless weather/satellite APIs, SQLite → the marginal cost per farmer conversation is fractions of a rupee.
- Same four-signal pattern generalizes to any rain-fed crop belt (Telangana, Karnataka, Bundelkhand) by swapping the district registry and crop calendar.

Deployment metric	Current build	State scale-up path
Districts monitored	16 (Vidarbha + Marathwada)	36 (all Maharashtra) — config only
Farmers reachable	every WhatsApp user in covered districts	WhatsApp Business API + Krishi Vibhag lists
Claim assistance	PMFBY letter auto-drafted in Marathi	direct PMFBY portal API integration
Officer surface	WhatsApp alerts + Streamlit heatmap	integration into existing Mahavedh/Krishi dashboards

8. Deployment Pathway & Sustainability

Designed for month two, not hour twelve (track slide 13):

- Ownership path: handoff to Krishi Vibhag (district agriculture offices) or an NGO operator; the officer alert flow mirrors their existing duty-roster practice.
- Low operating cost: one lightweight model file (LightGBM, <1 MB), free-tier inference, no GPU anywhere in the serving path.
- Maintainable by someone else: modular repo, one-command dataset regeneration and retraining, every data source documented, bias audit regenerates automatically.
- Pilot proposal: one monsoon season (Jun–Jan) across 3 talukas in Amravati district, measuring alert precision against officer-verified ground truth and PMFBY claim completion rates.

9. Live Demo (verified on a real phone)

The full loop below has already been executed end-to-end on a physical phone via the Twilio WhatsApp sandbox:

- 1. Farmer sends a Marathi voice note / text on WhatsApp.
- 2. ASR transcribes the note. Production path: Whisper large-v3 (hosted, free tier) — this is what runs in the demo. AI4Bharat IndicConformer is implemented as a configurable offline backend (ASR_BACKEND=ai4bharat) for the data-sovereign deployment described in §8; it is not the demo path.
- 3. Agent chains MCP tools: weather → mandi → NDVI → compute_fsi.
- 4. Calm day (live data, 5 July 2026): rainfall deficit 10.6%, NDVI 0.331, FSI 20.9 LOW → honest all-clear reply in Marathi, no alert. The system does not cry wolf.
- 5. Crisis scenario — a clearly labeled simulation replaying the Oct–Nov 2024 stress pattern (drought signal 0.70, price 18% below MSP): FSI 87.9 CRITICAL → officer WhatsApp alert + auto-drafted PMFBY claim letter in Marathi delivered back to the farmer. Every simulated message carries a visible SIMULATION tag; the deliberate contrast between step 4's real numbers and step 5's simulated ones is part of the demo.
- 6. Streamlit dashboard shows the Maharashtra FSI choropleth updating from the same cache.

Officer alert as delivered on WhatsApp: *"⚠ KisaanRaksha Alert: Amravati (Vidarbha) · FSI = 87.9 (CRITICAL) · Crop: cotton · Rainfall deficit: 57.4% · Market price ₹4,200/qtl (18% below MSP ₹7,710) ·*

NDVI 0.39 · Loan-repayment window: ACTIVE · Action: proactive outreach + PMFBY survey recommended · [SIMULATION — replays Oct-Nov 2024 Vidarbha stress pattern, not live data]"

Farmer-count integration (how many farmers are in the repayment window per taluka) requires the Krishi Vibhag crop-loan roster and is scoped as a pilot deliverable — the alert states only what the system has actually derived.

10. Team, Repository & Declaration

Item	Detail
Team	Sahil Karande · Pranav Shripannavar — 2 members
Repository	github.com/sahilkarande0918-cmd/Kisan-Raksha-project
Stack	Python · FastMCP · FastAPI · LightGBM · Fairlearn · ASR: Whisper large-v3 in production demo, AI4Bharat IndicConformer as implemented offline backend · Twilio WhatsApp · Streamlit · SQLite
Custom dataset	dataset/maharashtra_ag_stress_dataset.csv — flagged for IEEE Dataport
Bias audit	model/fairlearn_report.md — all gates pass
Declaration	All data sources public or synthetic; no real personal data collected; per Code of Conduct §8.3.

Technology must serve humanity — KisaanRaksha points it at the farmer the system forgot.